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FireCast: Leveraging Deep Learning to Predict Wildfire Spread

David Radke | Anna Hessler | Daniel Ellsworth | Environmental Science, Computer Science ·

International Joint Conference on Artificial… · 1 August 2019

TLDR FireCast predicts which areas surrounding a burning **wildfire** have high-risk of near-future **wildfire spread**, based on historical fire data and using modest computational resources, compared to a random **prediction** model and a commonly used **wildfire spread** model, Farsite. Expand

113 | PDF | Publisher | Save | Cite

Enhancing wildfire spread modelling by building a gridded fuel moisture content product with machine learning

Tyler C. McCandless | B. Kosović | William Petzke | Environmental Science, Computer Science ·

Machine Learning: Science and Technology · 20 August 2020

TLDR A high resolution, gridded, real-time FMC data sets that did not previously exist are developed for assimilation into operational wildland fire **prediction** systems based on ML and results show generally that the rule-based approaches have the lowest errors likely due to the sharp decision boundaries among the predictors. Expand

24 | PDF | IOP Publishing | Save | Cite

Impacts of estimated plume rise on PM2.5 exceedance prediction during extreme wildfire events: a comparison of three schemes (Briggs, Freitas, and Sofiev)

Yunyao Li | D. Tong | +12 authors | Fangjun Li | Environmental Science ·

Atmospheric Chemistry and Physics · 9 March 2023

Abstract. Plume height plays a vital role in **wildfire** smoke dispersion and the subsequent effects on air quality and human health. In this study, we assess the impact of different plume rise schemes… Expand

22 | [PDF] | PDF | Save | Cite

Recurrent Convolutional Deep Neural Networks for Modeling Time-Resolved Wildfire Spread Behavior

J. Burge | M. Bonanni | R. Hu | M. Ihme | Environmental Science, Computer Science ·

Fire technology · 28 October 2022

TLDR This work uses an autoregressive process in which a convolutional recurrent deep learning model makes **predictions** that propagate a **wildfire** over 15 min increments, and shows that even after 100 autoregressive **predictions** representing more than 24 h of simulated fire **spread**, the resulting models generate stable and realistic propagation dynamics. Expand

17 | [PDF] | Springer Nature | Save | Cite

Prediction of Wildfire Occurrence in the Southern Forest Regions of China in the Future Scenario

Jing Li | Duan Huang | +7 authors | Weike Li | Environmental Science, Geography · Forests ·

18 November 2024

In the context of global climate warming, climate change is subtly reshaping the patterns of **wildfires**. Therefore, it is particularly urgent to conduct in-depth research on climate change, **wildfires**,… Expand

1 | Publisher | Save | Cite

Wildfire Detection : Review of Emerging Technologies and Methodologies

Atharv Bhome · Environmental Science, Engineering · International Journal for Research in Applied… · 31 March 2025

Wildfires are uncontrolled fires that very quickly **spread** across forests, vegetation and grasslands powered or intensified by dry weather conditions, high temperatures and strong winds. These fires… Expand

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Early Adaptive Evaluation Scheme for Data-Driven Calibration in Forest Fire Spread Prediction

Edigley Fraga · A. Cortés · A. Cencerrado · P. Hernández · T. Margalef

Environmental Science, Computer Science · International Conference on Conceptual Structures · 25 May 2020

TLDR This paper proposes a new evaluation technique capable of reducing the overall calibration time by 60% when compared to the current data-driven approaches, and avoids wasting too much computing time running unfit individuals thanks to an early adaptive evaluation. Expand

4 PDF Publisher Save Cite

Systematic Literature Review on the Application of Mathematics, Statistics, and Computer Science in Wildfire Analysis

M. Najib · Sri Nurdiati · Environmental Science, Mathematics · Euler : Jurnal Ilmiah Matematika, Sains dan… · 28 April 2025

TLDR This study presents a systematic literature **review** of **wildfire** analysis techniques, focusing on trends from 2000 to 2025, and identifies the most widely applied methods, such as correlation, regression, classification, clustering, and artificial neural networks. Expand

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Predicting Wildfire Spread with Machine Learning and a Socio-Ecological Approach to Identify Areas of High Vulnerability

N. Shaik · Environmental Science, Computer Science · IEEE Conference on Technologies for… · 20 April 2025

TLDR This study marks the first comprehensive initiative across the USA to merge machine learning-driven **predictions** of **wildfire** intensity with social vulnerability assessments, determining areas in urgent need of support during **wildfire** disasters. Expand

IEEE Save Cite

A Neural Network Model of Wildfire Spread for Arbitrary Topography, Fuel Load Distribution, and Wind Field

Bryan Zhan · Environmental Science, Engineering · IEEE International Conference on Electro… · 29 May 2025

Due to climate change and other factors, **wildfires** have become more destructive and widespread. In January 2025, multiple fires ignited in Los Angeles, destroying 15,946 structures, damaging 1,953… Expand

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